

Exploratory Data Analysis (EDA) Midterm Proposal

Prepared for Prof. Vishwantha Rao, HCC ITAI 1371 Machine Learning

Prepared by FA26_ML_6171_16347 3_ITAI1371 Group

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This midterm proposal was prepared for ITAI 1371 Machine Learning (ML) by group 6171 16347 3. Our team has reviewed and all have approved this proposal for Prof. Rao's consideration.

GOAL 1: Reduce the risk of harm to minor children who use the Netflix streaming platform

GOAL 2: Reduce the potential of lawsuits against Netflix by preventing minor children from viewing mature, adult-oriented content (if we were employees of the company)

GOAL 3: Focus only on movies on the platform, eliminating TV shows from the dataset, and using rating as the feature to consider for publication to the Kids platform on Netflix.

GOAL 4: Demonstrate our understanding of the course material by completing every midterm requirement in a single, reproducible Jupyter/Colab notebook, showing that we can:

1. Load and inspect a raw dataset in Python (pandas) without ever editing the source file by hand.
2. Split data programmatically into 70% training / 30% testing sets, and perform EDA on the training set only.
3. Identify and fill NaN / null values with appropriate values (these will be our unrated <blank> ratings which will receive a 5 and prevent publication to the platform and trigger human review.
4. Scale data to change its range and normalize it to adjust its distribution.
5. Encode categorical data for type (movie or TV show), movie title, and rating, including one-hot encoding, into numeric form. Each movie title will have a unique numeric identifier. TV shows will all be eliminated from the dataset, and movie ratings will be given a unique numeric identifier as follows:
 - G – 0
 - PG – 1
 - PG-13 – 2
 - R – 3
 - NC17 – 4
 - <blank> - 5
6. Balance classes in the dataset (rating categories) so the model is not trained on a skewed distribution.
7. Engineer new features that support the ML problem we are solving. Features will be type, movie title, and rating. All other features (columns) in the dataset will be discarded.
8. Use Matplotlib to plot data distributions before and after each preprocessing step, and print sample records as they are processed.

9. Document our cleanup and preprocessing pipeline clearly enough to reuse the resulting dataset for the Finals ML models.

PROPOSAL

When logging into the Netflix streaming platform, options appear for user profiles and a special “kids” profile. Kids should be restricted from viewing mature, adult-oriented content with PG13, R, NC17 ratings as well as unrated content. ML could be used to automatically publish new content to be viewable by kids if and only if it is rated G or PG, provided it is trained correctly.

METHODOLOGY

1. Use the Netflix Movies and TV Shows dataset on Kaggle at <https://www.kaggle.com/datasets/shivamb/netflix-shows>. This dataset contains 8,807 rows and 12 columns, well above the assignment’s 2,000-row minimum, and will still comfortably exceed that minimum after TV shows are removed under Goal 3.
2. Split the dataset into 70% training and 30% test data.
3. Eliminate all TV shows from the dataset.
4. Determine which outliers exist which will be movies without ratings. These will require human review before publishing to the platform. These will be movies with <blank> ratings and numerically categorized as 5.
5. Train ML to publish content with G (0) and PG (1) ratings to be viewable by kids. Prohibit kids from viewing PG13 (2), R(3), NC-17(4), and unrated <blank> (5) content ratings.
6. Label unrated content as “human review required.”
7. Fill NaN and null values with appropriate value of “human review required”, scale numeric fields, normalize skewed distributions, and encode categorical fields such as rating, type, and title — all performed with Python code in Jupyter/Colab, never by manually editing the source file.
8. Check the rating distribution for class imbalance (we expect far fewer G/PG titles than mature ratings) and apply class balancing so the model isn’t biased toward the majority class.
9. Engineer a binary “kid-safe” target feature from the rating column (G=0, PG=1, PG13=2, R=3, NC17=4, Unrated <blank> = 5 for use in the Finals classification model.
10. Use Matplotlib to plot rating and feature distributions before and after each preprocessing step (e.g., class balance before/after), and print sample rows at each stage of the notebook.

On or before 03-Oct-2026, upload to GitHub each of the following deliverables:

- a. the dataset URL document,
- b. our PDF proposal,
- c. the Jupyter notebook (with the 70/30 split performed in Python and EDA limited to training data),
- d. the final cleaned dataset,
- e. and a reflection journal in which each team member describes their individual contribution.

NOTE: This proposal will serve in place of a contribution journal entry for the midterm.

EXPECTED RESULTS

We hypothesize the dataset will be substantially reduced by eliminating the TV shows and focusing on movies with ratings. We also anticipate there will be outliers in the data

without ratings which would need human review before publishing to the platform. We further expect the rating field to be imbalanced, with far more mature-rated titles than G/PG titles, which we will address with class balancing before training. We will use one-hot encoding to assign unique numerical values for each movie title, and a numerical value of 5 for unrated <blank> ratings which will prevent unrated movies from being published to the platform without human review.